

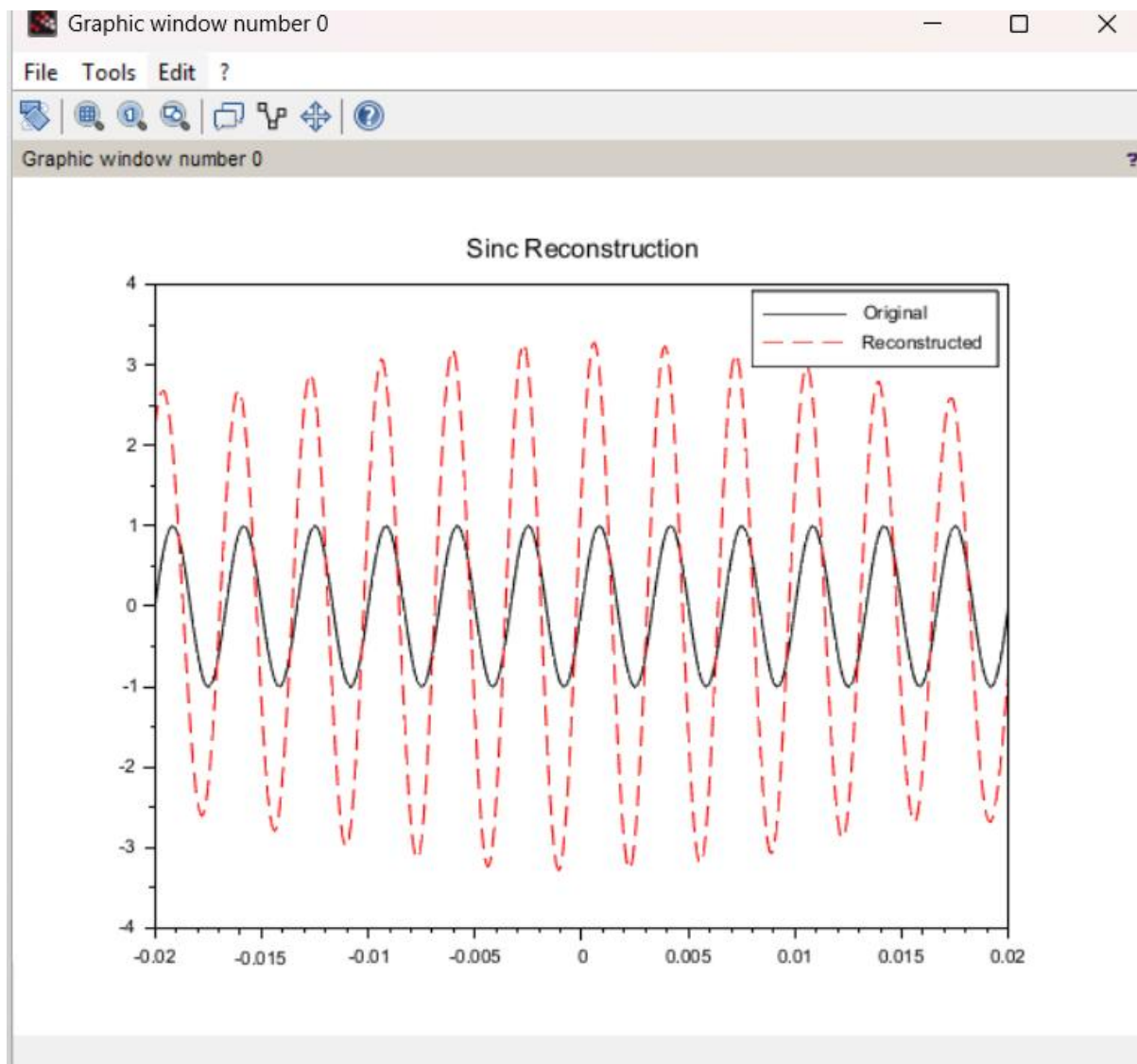
7. SIGNAL RECONSTRUCTION AND SPECTRAL ANALYSIS

Experiment 7.1

```
function y = sinc_interp(x, t, Ts)
y = 0*t;
for k = 1:length(x)
y = y + x(k) * sinc((t - (k-1 - length(x)/2)*Ts) / Ts);
end
endfunction

t = -0.02:1e-4:0.02;
f0 = 300;
x = sin(2*%pi*f0*t);

fs = 2000; Ts = 1/fs;
n = -40:40;
ns = n*Ts;
xs = sin(2*%pi*f0*ns);
x_rec = sinc_interp(xs, t, Ts);
clf;
plot(t, x, 'k');
plot(t, x_rec, 'r--');
legend("Original", "Reconstructed");
xtitle("Sinc Reconstruction");
```



Experiment 7.2clc;

clear;

clf;

//=====

// Fourier Series of Half-Wave Rectifier

//=====

// Parameters

A = 1;

```

f = 1;
T = 1/f;
dt = 0.01;
t = 0:dt:2*T;

// Input Signal
x_t = A * sin(2 * %pi * f * t);

// Half-Wave Rectified Signal
y_t = max(x_t, 0);

// Number of Harmonics
N_harmonics = 10;

//-----
// Fourier Series Coefficients
//-----

a0 = (2/T) * sum(y_t) * dt;

an = zeros(1, N_harmonics);
bn = zeros(1, N_harmonics);

for k = 1:N_harmonics
    an(k) = (2/T) * sum(y_t .* cos(2*%pi*k*f*t)) * dt;
    bn(k) = (2/T) * sum(y_t .* sin(2*%pi*k*f*t)) * dt;

```

```
end
```

```
//-----
```

```
// Signal Reconstruction
```

```
//-----
```

```
y_reconstructed = (a0/2) * ones(1, length(t));
```

```
for k = 1:N_harmonics
```

```
    y_reconstructed = y_reconstructed ...
```

```
        + an(k) * cos(2*%pi*k*f*t) ...
```

```
        + bn(k) * sin(2*%pi*k*f*t);
```

```
end
```

```
//-----
```

```
// Plotting
```

```
//-----
```

```
// Original Signal
```

```
subplot(3,1,1);
```

```
plot(t, y_t);
```

```
xlabel("Time (s)");
```

```
ylabel("Amplitude");
```

```
title("Half-Wave Rectified Output Signal");
```

```
// Reconstructed Signal
```

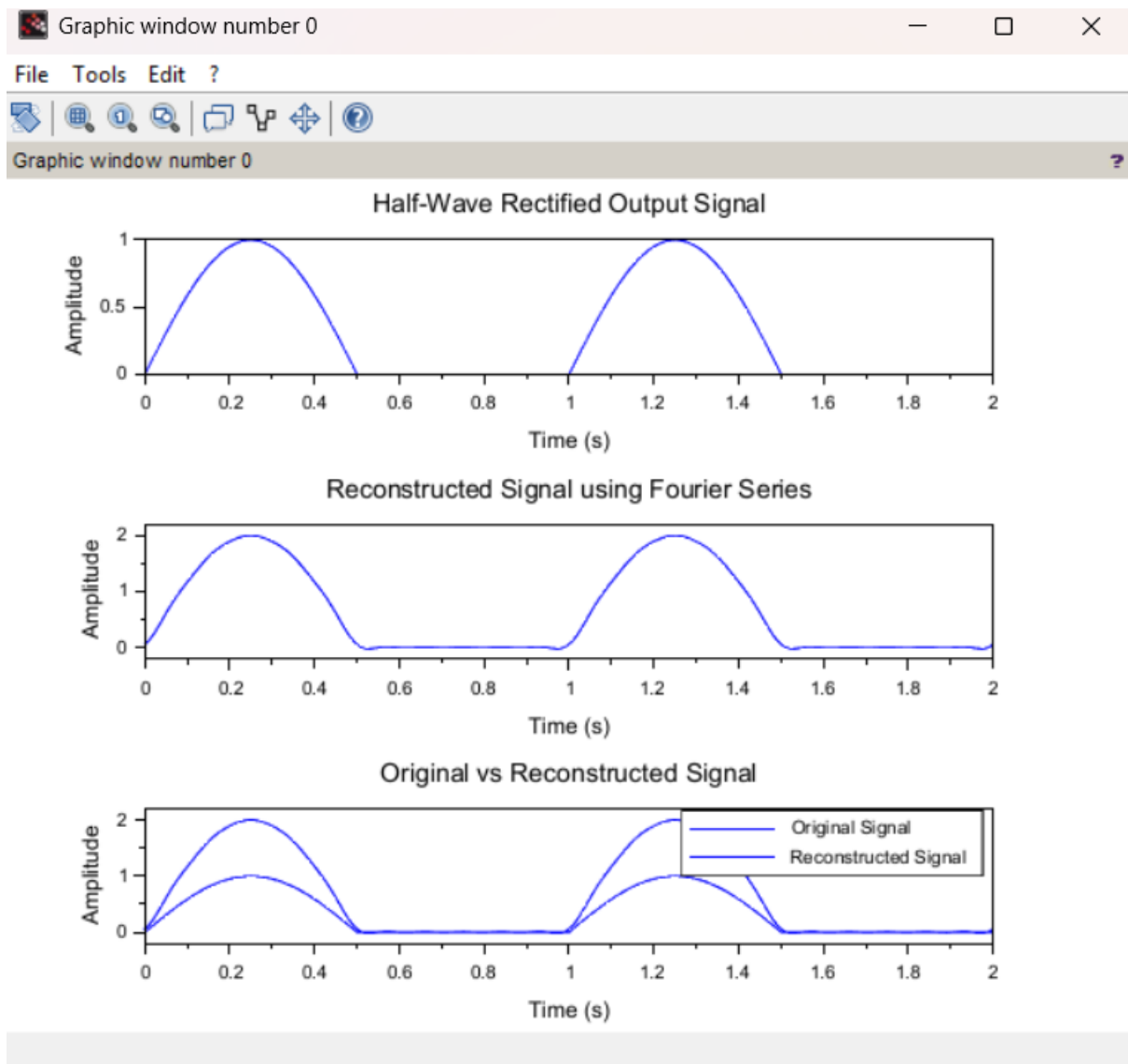
```
subplot(3,1,2);  
plot(t, y_reconstructed);  
xlabel("Time (s)");  
ylabel("Amplitude");  
title("Reconstructed Signal using Fourier Series");
```

```
// Comparison
```

```
subplot(3,1,3);  
plot(t, y_t);  
plot(t, y_reconstructed);
```

```
xlabel("Time (s)");  
ylabel("Amplitude");  
title("Original vs Reconstructed Signal");
```

```
legend("Original Signal", "Reconstructed Signal");
```



Experiment 7.3

// Time vector

t = -1:0.001:1; // fs = 1000 Hz

x = exp(-t.^2); // Gaussian pulse

// FFT length (must be >= length(x))

N = 2048; // choose bigger than length(x)

Xf = fft([x zeros(1, N - length(x))]); // zero-padding to N

// Frequency vector

fs = 1000; // Hz

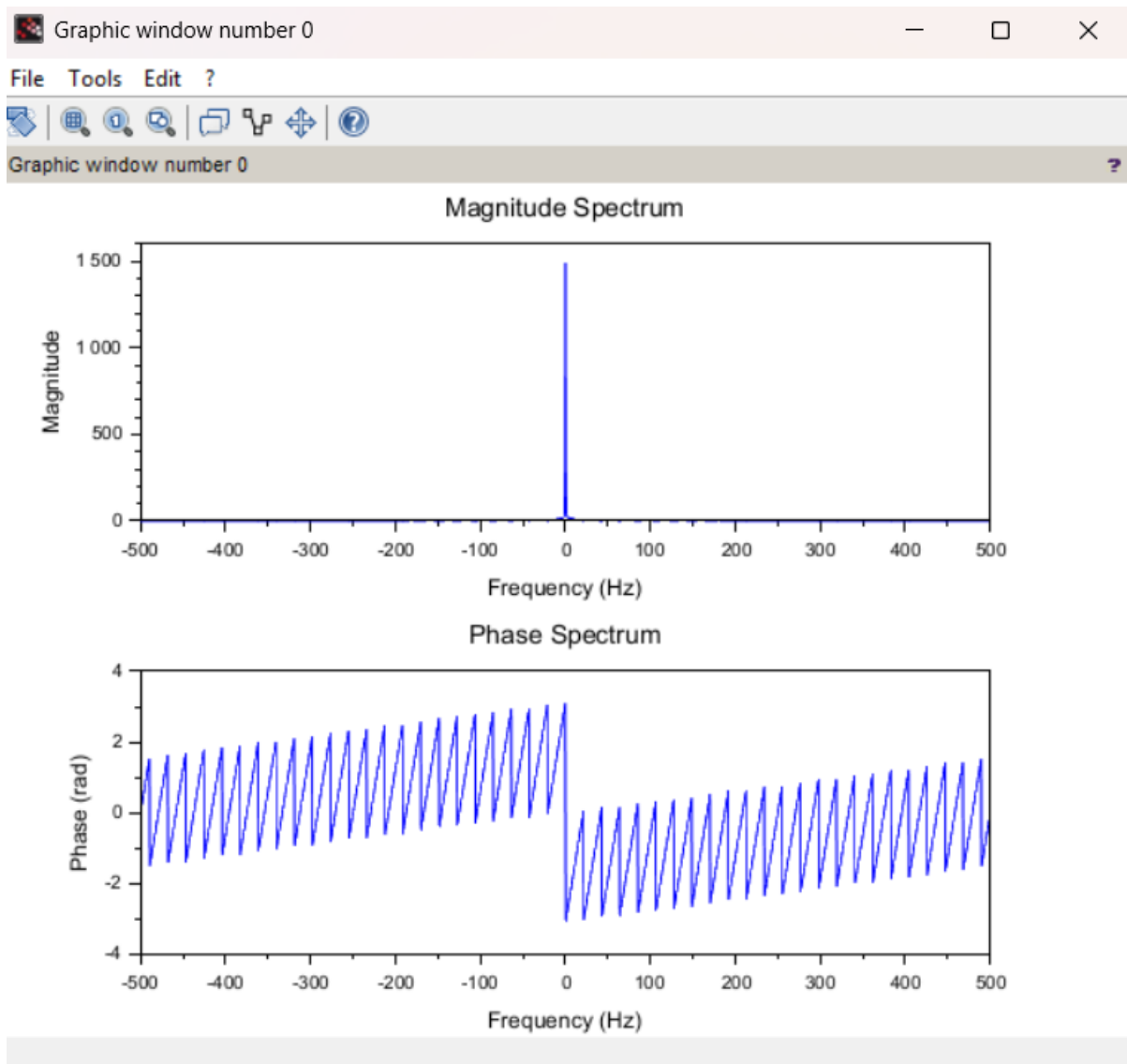
f = (-N/2:N/2-1) * (fs/N);

```
// Magnitude spectrum
magX = abs(fftshift(Xf));

// Phase spectrum
phaseX = atan(imag(fftshift(Xf)), real(fftshift(Xf)));

// Plot

clf;
subplot(2,1,1);
plot(f, magX);
xlabel("Magnitude Spectrum", "Frequency (Hz)", "Magnitude");
subplot(2,1,2);
plot(f, phaseX);
xlabel("Phase Spectrum", "Frequency (Hz)", "Phase (rad)");
```



Experiment 7.4

```
clc;
```

```
clear;
```

```
clf;
```

```
//=====
```

```
// Fourier Transform of a Rectangular Waveform
```

```
//=====
```

```
// Time Parameters
```



```
t_min = -5;
t_max = 5;
dt = 0.01;

t = t_min:dt:t_max;

// Rectangular Pulse
A = 1;
tau = 2;

x_t = A * (abs(t) <= tau/2);

//-----
// Fourier Transform using FFT
//-----

N = length(x_t);

X_f = fft(x_t, -1);

X_f_shifted = fftshift(X_f);

// Frequency Axis
f = (-N/2 : N/2-1) / (N*dt);

// Magnitude and Phase
```

```
magnitude_X_f = abs(X_f_shifted) / N;  
phase_X_f = atan(imag(X_f_shifted), real(X_f_shifted));
```

```
//-----  
// Plotting  
//-----
```

```
// Rectangular Pulse  
subplot(3,1,1);  
plot(t, x_t);  
xlabel("Time (s)");  
ylabel("Amplitude");  
title("Rectangular Waveform");
```

```
// Magnitude Spectrum  
subplot(3,1,2);  
plot(f, magnitude_X_f);  
xlabel("Frequency (Hz)");  
ylabel("|X(f)|");  
title("Magnitude of Fourier Transform");
```

```
// Phase Spectrum  
subplot(3,1,3);  
plot(f, phase_X_f);  
xlabel("Frequency (Hz)");  
ylabel("Phase (rad)");
```

```
title("Phase of Fourier Transform");
```

